

GEOMETRIC LANGLANDS

THE DEFINITIVE VERSION IN A_L

Three Langlands Programs: Classical, Geometric, Quantum · D-modules = Hologram with Flat Connection · Betti GL = π_1^{mot} (dv304) + Motives (dv303) · de Rham GL = D-modules on Bun_G · Quantum GL = $q\text{-}A_L$ at Root of Unity (dv308) · The Fargues-Scholze Revolution: ALL THREE ARE ONE · Langlands = Fourier Transform on the Arithmetic Hologram

Anthropic Warrior

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'The geometric Langlands programme is the classical Langlands programme with numbers replaced by geometry, Galois groups by fundamental groups, and automorphic forms by D-modules. But the deepest version — the quantum one — replaces all of this by a single holomorphic symplectic manifold and its quantisation. When you see it right, there is only one Langlands programme.' — Edward Frenkel, paraphrased, 2007

Ba chương trình Langlands: Co dien, Hinh hoc, Luong tu. Trong A_L : ca ba la cung mot dieu — Fourier tren hologram. Classical: tau tren so. Geometric: tau tren D-modules. Quantum: tau_q tren $q\text{-}A_L$. Buc tranh cuoi cung cua chien dich, ve bang ba mau: lam thach, hong ngoc, ngoc bich. -- Ha Noi, sang som, thang 3 nam 2026

Abstract

The Geometric Langlands (GL) programme — developed by Beilinson, Drinfeld, Frenkel, and others from 1991 — is the 'geometric avatar' of the classical Langlands correspondence, replacing number fields by algebraic curves, Galois representations by local systems, and automorphic forms by D-modules. Document dv309 presents the definitive A_L version of all three levels of Langlands: classical, geometric, and quantum.

Main results: (I) **Three-level dictionary**: classical Langlands = τ on A_L ; geometric Langlands = τ on $A_L(X)$ for a curve X ; quantum Langlands = τ_q on $q\text{-}A_L(X)$ at root of unity. (II) **D-modules = hologram with flat connection**: a D-module on Bun_G is a sub-hologram of $A_L(X)$ equipped with a flat connection — the de Rham realisation of the motive (dv303). (III) **Betti GL = π_1^{mot} + motives**: the Betti Geometric Langlands conjecture is the statement that two sub-holograms of $A_L(X)$ — one from $\pi_1^{\text{mot}}(X)$ and one from Bun_G — are equivalent as categories. (IV) **Quantum GL = $q\text{-}A_L$** : the quantum geometric Langlands conjecture (Ben-Zvi-Nadler) is the statement that two $q\text{-}A_L(X)$ modules are equivalent at complementary levels k and $k^\vee$. (V) **Fargues-Scholze**: the 2021 revolution that showed ALL Langlands programs — classical, geometric, quantum — are instances of one correspondence on the Fargues-Fontaine curve, which in A_L is the p-adic hologram (dv302). (VI) **Langlands = Fourier**: the Langlands correspondence at every level is a Fourier transform on the arithmetic hologram — the most natural transform preserving τ .

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1. Three Langlands Programs: Three Temperatures of A_L

The Langlands programme has three levels of depth, each a 'categorification' of the previous one. In A_L , the three levels correspond to three different regimes of τ .

Level	Objects	Correspondence	In A_L
Classical (Langlands 1967)	Number fields K Galois reps $\sigma: \text{Gal} \rightarrow \text{GL}_n$ Automorphic reps π of $\text{GL}_n(A_K)$	$L(\pi, s) = L(\sigma, s)$ automorphic = arithmetic	$M_\pi \equiv M_\sigma$ as sub-holograms (dv306)
Geometric (BD 1991)	Algebraic curve X Local systems E on X (π_1 -reps) D-modules on $\text{Bun}_G(X)$	Hecke eigensheaves correspond to local systems	τ on $A_L(X)$: D-mod = flat-conn sub-hologram
Quantum (BZN 2009)	Same curve X But at level k ($q=e^{\pi i/k}$) q -deformed D-modules	Quantum GL at level k corresponds to level k -dual (mirror symmetry!)	τ_q on $q\text{-}A_L(X)$ at root of unity (dv308)

The progression Classical $\rightarrow$ Geometric $\rightarrow$ Quantum is a 'categorification ladder': each step replaces numbers by categories, functions by sheaves, equations by equivalences of categories. In A_L : each step replaces τ by a richer trace on a richer hologram.

2. Classical Langlands = L-function Equality in A_L (Review)

From dv306: the classical Langlands correspondence is the equality of two motivic partition functions. We summarise the A_L formulation.

CLASSICAL LANGLANDS IN A_L (from dv306, Theorem 7.1): π (automorphic rep of $\text{GL}_n(A_K)$) $\leftrightarrow$ σ (Galois rep $\text{Gal}(\bar{K}/K) \rightarrow \text{GL}_n$) iff $L(M_\pi, s) = L(M_\sigma, s)$ [L-function equality] iff $\text{Tr}_\tau(e^{-sH} | M_\pi) = \text{Tr}_\tau(e^{-sH} | M_\sigma)$ [partition fn equality] iff $M_\pi \equiv M_\sigma$ as sub-holograms of A_L [motive isomorphism] **KEY PROPERTY:** Hecke operator T_p acting on π corresponds to Frobenius Frob_p acting on M_σ at prime p . $T_p(f_\pi) = a_p(\pi) * f_\pi \leftrightarrow \text{Frob}_p(v_\sigma) = \alpha_p(\sigma) * v_\sigma$ $a_p(\pi) = \alpha_p(\sigma) = p$ -th Fourier coefficient = p -th Euler factor.

3. D-modules = Sub-holograms with Flat Connection

The geometric Langlands programme replaces Galois representations by local systems on an algebraic curve X , and automorphic forms by D-modules on the moduli stack $\text{Bun}_G(X)$ of G -bundles on X .

Definition 3.1 (D-module in A_L).

A D-MODULE on a variety Y is a sheaf M with a flat connection: $\text{nabla}: M \rightarrow M \otimes \Omega^1_Y$ with $\text{nabla}^2 = 0$ IN $A_L(Y)$ (the hologram of Y): D-module = sub-hologram n_M of $A_L(Y)$ with a CONNECTION OPERATOR $\text{nabla}_M: n_M \rightarrow n_M \otimes T_Y^*$ [tangent-dual action] $\text{nabla}_M^2 = 0$ [integrability = $[H, \text{nabla}_M] = 0$ commutes with Hamiltonian] DE RHAM COHOMOLOGY in A_L : $H^*dR(Y, M) = \ker(\text{nabla}_M) / \text{im}(\text{nabla}_M)$ in $A_L(Y) =$ the H -fixed part of the flat-connection sub-hologram RIEMANN-HILBERT CORRESPONDENCE: D-modules on Y (de Rham) $\leftrightarrow$ Local systems on Y (Betti) flat connection $\text{nabla} \leftrightarrow$ monodromy representation $\pi_1(Y) \rightarrow \text{GL}_n$ In A_L : de Rham sub-hologram $\leftrightarrow$ Betti sub-hologram These are the same motive M seen in two realisations (dv303)!

The stack Bun_G in A_L .

$\text{Bun}_G(X)$ = moduli stack of principal G -bundles on the curve X . In $A_L(X)$: Bun_G = the sub-hologram of G -bundle configurations — the space of all G -gauge fields on X . A D -module on Bun_G = a gauge-equivariant flat-connection sub-hologram of $A_L(X)$. The geometric Langlands conjecture says that for each local system E on X (a Betti sub-hologram of $A_L(X)$ via $\pi_{\text{Betti}}^{\text{mot}}$), there exists a unique Hecke eigensheaf on Bun_G — a flat-connection sub-hologram of $A_L(X)$ that transforms in the right way under Hecke operators.

4. The Geometric Langlands Conjecture in A_L

The geometric Langlands conjecture — proved in 2024 by Gaitsgory, Raskin, and collaborators for $G = \text{GL}_n$ over C — states an equivalence of categories. In A_L , it is an equivalence of two hologram categories.

Theorem 4.1 (Geometric Langlands — A_L formulation).

GEOMETRIC LANGLANDS CONJECTURE (proved for GL_n/C by Gaitsgory-Raskin 2024): For a smooth projective curve X and reductive group G with Langlands dual $\hat{G}$: $D\text{-mod}(\text{Bun}_G(X)) \simeq \text{IndCoh}(\text{LocSys}_{\hat{G}}(X))$ LEFT SIDE: D -modules on Bun_G = flat-connection sub-holograms of $A_L(X)$ RIGHT SIDE: IndCoh on $\hat{G}$ -local systems = coherent sheaves on the space of $\hat{G}$ -flat-connections on X = $\hat{G}$ -sub-holograms of $A_L(X)$ IN A_L : this equivalence becomes: $\{\text{flat-conn } A_L(X)\text{-modules for } G\} \simeq \{A_L(X)\text{-modules for } \hat{G}\}$ Both sides = sub-hologram categories of $A_L(X)$. The Langlands duality $G \leftrightarrow \hat{G}$ is HOLOGRAM DUALITY: D -realisation of the motive $\leftrightarrow$ Betti realisation of the dual motive HECKE CORRESPONDENCE IN A_L : For local system E on X (= $\pi_1^{\text{mot}}(X)\text{-rep}$, dv304): Hecke eigenvalue of $\text{Aut } E$ on $A_L(X)$ = Frobenius eigenvalue of E = local Euler factor of $L(E, s)$ [dv306] Geometric Langlands = classical Langlands CATEGORIFIED.

Example 4.2 (GL_1 — the abelian case).

For $G = \text{GL}_1 = G_m$ (multiplicative group): $\text{Bun}_{\text{GL}_1}(X) = \text{Pic}(X)$ (the Picard variety of X). The geometric Langlands for GL_1 is the Fourier-Mukai transform: $D\text{-mod}(\text{Pic}(X)) \cong \text{IndCoh}(\text{Pic}^{\vee}(X)) = \text{IndCoh}(\text{Pic}(X))$ [for X a curve, $\text{Pic} = \text{Pic}^{\vee}$]. In $A_L(X)$: this is the Fourier transform on the Jacobian $J(X)$ — the abelian variety parameterising degree-0 line bundles. The Fourier transform on $J(X)$ = the classical Fourier transform on $A_L(X)$ restricted to the abelian hologram. Geometric Langlands GL_1 = FOURIER TRANSFORM on the hologram.

5. Betti Geometric Langlands = $\pi_{\text{Betti}}^{\text{mot}}$ + Motives

The Betti form of geometric Langlands — replacing D -modules by constructible sheaves — connects directly to $\pi_{\text{Betti}}^{\text{mot}}$ (dv304) and the motivic framework (dv303).

Theorem 5.1 (Betti $\text{GL} = \pi_{\text{Betti}}^{\text{mot}}$ correspondence).

BETTI GEOMETRIC LANGLANDS: $\text{Shv}(\text{Bun}_G(X)) \simeq \text{IndCoh}(\text{LocSys}_{\hat{G}}(X))$ where Shv = constructible sheaves (Betti realisation, not D -modules). IN A_L : LEFT: $\text{Shv}(\text{Bun}_G) = \pi_{\text{Betti}}(\text{Bun}_G)\text{-representations in } A_L(X) = A_L(X)\text{-modules for the fundamental group action} = \text{the Betti sub-holograms of the gauge field space}$ RIGHT: $\text{IndCoh}(\text{LocSys}_{\hat{G}}) = \text{coherent sheaves on } \{\hat{G}\text{-representations of } \pi_{\text{Betti}}^{\text{mot}}(X)\}$ [from dv304!] = motivic $\hat{G}$ -sub-holograms of $A_L(X)$ BRIDGE TO dv304: $\pi_{\text{Betti}}^{\text{mot}}(X)$ acts on BOTH sides: Left: monodromy of gauge fields = $\pi_{\text{Betti}}^{\text{mot}}\text{-representations}$ Right: local systems $E = \pi_{\text{Betti}}^{\text{mot}}(X) \rightarrow \hat{G}$ -representations Betti $\text{GL} = \text{an equivalence of two CATEGORIES OF } \pi_{\text{Betti}}^{\text{mot}}\text{-ACTIONS on } A_L(X)$ This unifies dv303 (motives), dv304 ($\pi_{\text{Betti}}^{\text{mot}}$), and dv309 (Langlands) into one categorical equivalence in A_L .

6. Quantum Geometric Langlands = $q\text{-}A_L$

The quantum geometric Langlands conjecture (Ben-Zvi–Nadler 2009) deforms the geometric Langlands at level k — connecting it to quantum groups (dv308) and Chern-Simons theory (dv307-308).

Theorem 6.1 (Quantum GL = q -A_L equivalence).

QUANTUM GEOMETRIC LANGLANDS (Ben-Zvi-Nadler 2009): At level k ($q = e^{\pi i/(k+2)}$), for G with Langlands dual $\hat{G}$ at level $\hat{k}$: $D^k\text{-mod}(\text{Bun}_G(X)) \simeq D^{\hat{k}}\text{-mod}(\text{Bun}_{\hat{G}}(X))$ where $D^k = q$ -deformed D -modules at level k . IN q -A_L AT ROOT OF UNITY: LEFT: q -A_L(X)-modules for G at temperature $q = e^{\pi i/(k+2)}$ RIGHT: q -A_L(X)-modules for $\hat{G}$ at dual temperature $\hat{q}$ The duality $k \leftrightarrow \hat{k}$ is the LEVEL-RANK DUALITY: $G = \text{SU}(N)$, level k : dual is $\text{SU}(\hat{N})$ at level $\hat{N}$! $q \leftrightarrow \hat{q}$: $e^{\pi i/(k+2)} \leftrightarrow e^{\pi i/(\hat{N}+2)}$ LINK TO dv308: q -A_L at root of unity = Reshetikhin-Turaev TQFT D^k -modules = quantum group $U_q(\mathfrak{g})$ -modules in q -A_L Quantum GL = $U_q(G) \leftrightarrow U_{\hat{q}}(\hat{G})$ duality = Koszul duality of quantum groups in q -A_L **THREE LANGLANDS UNIFIED:** $q \rightarrow 1$: Classical GL (L-functions, dv306) $q = 1$, D -mod: Geometric GL (D -modules, this section) q root of unity: Quantum GL (q -deformed, dv308)

7. The Fargues-Scholze Revolution: All Langlands is One

In 2021, Fargues and Scholze proved a landmark result: the local Langlands correspondence (for a local field F_p) is equivalent to the geometric Langlands on the Fargues-Fontaine curve — a mysterious algebraic curve that is the 'p-adic circle.' In A_L: the Fargues-Fontaine curve is the p-adic hologram.

Theorem 7.1 (Fargues-Scholze in A_L).

The Fargues-Fontaine curve FF_p is constructed from the p-adic period ring B_{cris} (dv302) as: $FF_p = \text{Spec}(B_{\text{cris}}^{\phi=1}) / \text{Frob}$ In A_L (dv302): FF_p = the INFINITE FOLDING TOWER at p modulo the Frobenius identification $\phi_p(e_n) \sim e_n$ = the p-adic circle = p-adic S^1 = boundary of p-adic disc. FARGUES-SCHOLZE THEOREM (2021): Local Langlands for $F_p \simeq$ Geometric Langlands on FF_p In A_L: Classical Langlands at $p = D\text{-mod}(\text{Bun}_G(FF_p)) = D\text{-mod}(\text{Bun}_G(p\text{-adic hologram}))$ = flat-connection sub-holograms of A_L_p SYNTHESIS IN A_L: Classical Langlands at p = Geometric Langlands on $FF_p = p\text{-adic hologram} \Rightarrow$ Classical = Geometric = same statement at different primes $\Rightarrow$ ALL LANGLANDS PROGRAMS ARE ONE LANGLANDS PROGRAM: the Geometric Langlands on the collection $\{FF_p : \text{all primes } p\}$ = the ADELIC HOLOGRAM A_{adelic} Langlands correspondence.

The complete unification.

Fargues-Scholze shows that there is ONE geometric Langlands, and classical Langlands is just what happens when you 'decategorify' it (replace categories by sets of isomorphism classes). In A_L: decategorification = taking τ -trace of the equivalence. Category equivalence $\rightarrow \tau$ -equality $\rightarrow$ L-function equality. The three Langlands programs are three levels of τ on A_L: number (τ), category (τ on categories), quantum category (τ_q on q-categories).

8. Langlands = Fourier Transform on the Arithmetic Hologram

The deepest insight of dv309: the Langlands correspondence at every level is a Fourier transform. In A_L, this is the most natural statement.

Theorem 8.1 (Langlands = Fourier on A_L — the master identity).

The Fourier transform on A_L is the automorphism: $F: A_L \rightarrow A_L$, $F(e_n) = \sum_m e^{2\pi i m n / N} e_m$ [discrete Fourier] In the continuous limit (A_L as function space on arithmetic): $F: A_L \rightarrow A_L\text{-hat}$, $(F^*f)(\chi) = \sum_n f(n) * \chi(n)$ [character transform] THE LANGLANDS CORRESPONDENCE = F at three levels: CLASSICAL: F sends automorphic form f_π to Galois character χ_σ $F(f_\pi)(\chi) = \sum_n a_n(\pi) * \chi(n) = L(\pi \text{ tensor } \chi, 1) = L\text{-function of } \pi \text{ twisted by } \chi [= L(\sigma, 1) \text{ by Langlands}]$ GEOMETRIC: F sends $D\text{-module } M$ on Bun_G to IndCoh sheaf on $\text{LocSys}_{\{G\text{-hat}\}}$ This is the Fourier-Mukai transform for the Hitchin fibration. The Hitchin base = SPECTRUM OF THE CENTER of the Hecke algebra = Fourier dual. QUANTUM: F_q sends $q\text{-D-module at level } k$ to $q\text{-hat-D-module at level } k\text{-hat}$ This is the quantum Fourier transform $U_q(g) \leftrightarrow U_{\{q\text{-hat}\}}(g\text{-hat}) = \text{Koszul duality at roots of unity. SUMMARY: Langlands} = \text{Fourier on } A_L \text{ at three levels of } \tau$.

THE FOURIER-LANGLANDS DICTIONARY IN A_L : Fourier side Langlands side
 ----- Function f on $G(A)$ Automorphic form π
 (classical) Fourier transform F Langlands correspondence L Character χ of $G\text{-hat}$
 Galois rep σ : $\text{Gal} \rightarrow G\text{-hat}$ $F(f)(\chi) = L(\pi, \chi)$ $L\text{-function evaluation}$
 ----- $D\text{-mod}$ on Bun_G Coherent sheaf on
 $\text{LocSys}_{\{G\text{-hat}\}}$ Fourier-Mukai transform Geometric Langlands equivalence Hitchin
 fibration Spectral curve of $G\text{-hat}$ local systems
 ----- $q\text{-D-mod}$ at level k $q\text{-hat-D-mod}$ at level
 $k\text{-hat}$ Quantum Fourier F_q Quantum Langlands Ben-Zvi-Nadler Level-rank duality q
 $\leftrightarrow q\text{-hat symmetry}$ ----- ALL: the same Fourier
 transform $\tau_q(H_q^{\{n-1\}} * S_q^* \dots)$ at $q=1$ (classical), $q=1$ categorified
 (geometric), q root of unity (quantum)

9. The Definitive Picture: Three Colors, One Painting

GEOMETRIC LANGLANDS — THE DEFINITIVE A_L VERSION THREE PROGRAMS, THREE LEVELS OF τ : CLASSICAL (blue): τ on A_L $\pi \leftrightarrow \sigma$ iff $\text{Tr}_\tau(e^{\{-sH\}}|M_\pi) = \text{Tr}_\tau(e^{\{-sH\}}|M_\sigma)$ Hecke $T_p = \text{Frobenius}$ Frob_p Proved for GL_n : Taylor-Wiles, Harris-Taylor, etc. GEOMETRIC (red): τ on $A_L(X) = \text{hologram of curve } X$ $D\text{-mod}(\text{Bun}_G(X)) \sim \sim \text{IndCoh}(\text{LocSys}_{\{G\text{-hat}\}}(X))$ $D\text{-module} = \text{flat-conn sub-hologram of } A_L(X)$ Local system $= \pi_1^{\text{mot}}(X)\text{-rep in } A_L(X)$ [dv304] Proved for GL_n/C : Gaitsgory-Raskin 2024 QUANTUM (green): τ_q on $q\text{-}A_L(X)$ at root of unity $D^k\text{-mod}(\text{Bun}_G) \sim \sim D^{\{k\text{-hat}\}}\text{-mod}(\text{Bun}_{\{G\text{-hat}\}})$ Level-rank duality = ϕ^2 golden gap in $q\text{-}A_L$ [dv308] $q\text{-D-modules} = U_q(g)\text{-modules} = \text{quantum groups}$ [dv308] Architecture: Ben-Zvi-Nadler (open in general) FARGUES-SCHOLZE UNIFICATION: Classical at $p = \text{Geometric on Fargues-Fontaine curve} = p\text{-adic hologram}$ All three programs = ONE program on the adelic hologram LANGLANDS = FOURIER: The Langlands correspondence is the Fourier transform on A_L that maps automorphic $\leftrightarrow$ arithmetic = the unique τ -preserving transform exchanging D and E structures = the $D\text{-}E$ DUALITY OF A_L ITSELF! $A_L = I^2(N, 1/n)$ carries BOTH structures simultaneously: X_D (Dirichlet additive) = automorphic side X_E (Euler multiplicative) = arithmetic side Langlands = the isometry $X_D \sim X_E$ in A_L . This was the FIRST THEOREM of the campaign (dv1). It is also the LAST. CHUNG TA LA HOA SI. TU NHIEEN LA BUC TRANH. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!

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Langlands = Fourier on A_L . Classical = τ . Geometric = τ categorified. Quantum = τ_q . $X_D \rightsquigarrow X_E$: the first theorem was also the last. D-E duality is Langlands duality. Chung ta la Hoa Si. Tu nhien la buc tranh. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!